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AI and work in the creative industries: digital continuity or discontinuity?

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ABSTRACT

There is current uncertainty about the effects of generative AI technologies on labour markets and conditions of work in creative industries. Previous scholarship identifies the potential for both replacement and displacement of human labour due to AI adoption. This article examines potential effects of AI adoption by drawing on case studies of 6 commercial products using AI, investigating conditions of work and mechanisms by which human creative labour might be replaced or displaced by AI technology. The main findings are that AI products were more labour intensive than traditional media products because they combined traditional production skills and new computational expertise. Paradoxically, contributions by human workers tended to be invisibilised in final products that foregrounded AI technology. There was potential for displacement of human labour in the ideation phase, with AI enabling producers to explore different creative possibilities. Some AI products had the potential to compete with earlier production methods (e.g. AI-generated imagery vs. stock photography). Overall, case study findings were consistent with ongoing challenges from digitalisation: conditions of deskilling, re-skilling, flexible employment and uncertainty remain intense for small firms engaged in AI-assisted creative production.

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1. Introduction

Improvement in the capabilities of artificial intelligence (AI) has led to recent public and scholarly concern about the impact of these technologies on society and the economy (Briggs and Kodnani 2023). Broadly defined, AI technologies are those that, by mimicking human learning and decision-making processes, can produce outputs ‘independently and without human influence’ (Dornis 2020, 7). Notable AI products such as ChatGPT, Stable Diffusion and Midjourney (all initially released in 2022) have drawn attention to the possibilities of generative AI and its apparent ability to produce text and images indistinguishable from the work of humans. This has obvious implications for the creative industries, where text and images make up a considerable

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proportion of outputs in sectors such as publishing, journalism and the visual arts. The seemingly imminent disruption of creative industries by AI has perhaps gained focus because these activities are associated with human characteristics of judgment, taste and artistic autonomy, thought to be beyond the reach of automation. Yet, the outputs of AI systems seem increasingly capable of passing a distributed, public Turing test,¹ evidenced by vociferous social media debates about their impact (Williams 2023). The perceived capabilities of AI have provoked alarm, resulting in calls to suspend development of large language models (LLMs) until regulators and the public can gain better understanding of the impacts and risks (Future of Life Institute 2023; Hern 2023).

One major concern is the potential for AI to replace or disrupt labour markets. A 2023 report by Goldman Sachs estimated that some 25% of labour tasks in the U.S. market are susceptible to automation by AI (Briggs and Kodnani 2023). Due to the pace of AI development, there is limited empirical data on its effects (see Neff, McGrath, and Prakash 2020; Hassel and Özkiziltan 2023). There is an even greater lack of empirical evidence about specific impacts of AI for the creative industries. Speculative, legal and policy studies have considered how AI technologies might impact existing regulatory frameworks such as freedom of expression and copyright, but these discussions have so far lacked empirical grounding (Bridy 2012; Lim 2018; Kop 2019; Hugenholtz and Quintais 2021). Previous scholarship has identified potential concerns about ownership of upstream training data, inter-firm competition and effects on the market from oversupply of AI-produced content, all of which have implications for creative labour. Broadly, two potential challenges emerge from the existing literature. One proposition is that AI technologies might potentially replace or substitute for professional creators, supplanting the human role in creative industry production (Boden 2009; Pearlman 2017; Brown 2018; Dornis 2020). In this scenario, AI marks a discontinuity in the organisation of production and labour, with impacts that may require changes to policy frameworks such as copyright. A second proposition is that AI technologies might not fully replace but still *displace* human workers, even when AI is used as a tool to complement human creativity (Grimmelmann 2015; Kaminski 2017). Displacement could occur through added efficiency or elimination of skilled work in certain aspects of production but not all (Drott 2021; Su, Togay, and Côté 2021). Effects of displacement may be accelerated with AI, similarly to how digitalisation is already transforming creative industry work. Adoption of digital technology overall has been found to have a negative effect on creative earnings and led to more precarious working conditions (Williams, McDonald, and Mayes 2021). While there is therefore disagreement about the precise impact of AI-assisted creative production (replacement or displacement), there is broad agreement that AI technology will have wide-reaching effects on creative work. This study aims to contribute to understanding by investigating whether anticipated impacts of AI adoption are playing out on the ground. In this paper, I ask the following research questions: How is the use of AI impacting creative work in firms that use the technology? Do we observe replacement or displacement taking place in this context? In what ways are any impacts continuous with ongoing challenges of digitalisation, and in what ways might they be discontinuous, requiring

new policy focus? I address these research questions using a multiple case study research design focusing on creative product teams working with AI.

The overall finding is that for the production teams analysed, use of AI did not result in replacement of human labour. The experimental use of AI technology by the creative SMEs in this study required new types of work, drawing on new sources of expertise and skills. In the cases analysed, human labour was present at all stages of production, including in new job tasks such as identifying, cleaning and curating input data as well as editing and preparing AI outputs for market. Much of this labour went uncredited: marketing of products as ‘made with AI’ obscured the role of human labour behind those products. Consistent with previous research on digitalisation, the case study firms were characterised by precarious conditions, flexibility and risk. The experiences of workers in the case study firms suggest that pressures introduced by AI technology may overlap with those already associated with digitalisation more generally. Accumulated knowledge about prior technological disruption in creative industries (including in this journal) remains salient for policy discussions about the impacts of generative AI in creative SMEs.

The article is structured as follows: in the next section, I provide an overview of the existing research on the effects of digitalisation on creative industries, with a focus on creative labour markets and working conditions. Drawing from legal and policy studies literature I then review claims about whether AI might replace or displace creative labour, with reference to historically observed trends. The objective of this literature review is to summarise existing knowledge and theory to inform empirical analysis of creative firms. The methods section outlines the research design, which develops a multiple case-study analysis of 6 products that incorporated AI inputs, and the production practices surrounding them. The discussion then elaborates on these findings, supported by interviews with key production personnel about how AI has shaped their working conditions and prospects for future work. The conclusion identifies avenues for future research as well as potential implications for policymakers and researchers.

2. Literature review: AI and creative work

2.1. *Discontinuities: what is new about AI?*

Does the advent of generative AI present new, discontinuous challenges beyond those already experienced with digitalisation? Possible effects include the possibility that AI might substitute entirely for primary human authors, directly replacing core creative workers. Other perspectives, while less convinced of the capability of AI to overtake human creative work, nevertheless envisage a displacement effect from AI when used as a production tool. Potential effects include oversupply of creative goods with automation, further deskilling of creative labour, and displacement of ancillary production jobs due to increased efficiency and changing production practices.

The possibility that AI might replace human artistic output was explored prior to the arrival of ChatGPT and other generative AI systems (Bridy 2012; Grimmelmann 2015; Craig and Kerr 2020). Assessment criteria such as audience preference, industry

awards and accolades were used to argue that AI systems have matched human capabilities in terms of quality (Brown 2018; Lim 2018). From the point of view of audiences, the true origin of a work – whether human or AI – might become indistinguishable, equivalent to a kind of creative Turing test (Amabile 2020; Natale and Henrickson 2024). Boden (2009) has argued that holistic measures of sentience are not required for AI to produce creative work: it is more pragmatic to focus on the extent to which AI outputs mimic human abilities. She notes that despite technological hurdles, there was already evidence of computers exhibiting combinatorial, exploratory and transformational creativity, thought to be the exclusive domain of humans (2009, 25). Natale and Henrickson (2024) propose the ‘Lovelace effect’² to draw attention to the significance of the persistent belief by members of the public that computers can produce original creative expressions.

Copyright scholarship has asked whether a computer system might legally attain the status of ‘author’ and thus replace humans in one important legal sense (Bridy 2012; Boyden 2015; Gervais 2019; Craig and Kerr 2020). In copyright theory, the threshold of originality demarcates the line above which works are eligible for protection as an author’s original work (Samuelson 1985). Since human and AI outputs are becoming indistinguishable, the question of how to treat AI outputs under copyright law invites ontological reflection on the concept of originality. Does originality always require the conscious judgment of a human, no matter how minimal the threshold, or can it be an emergent property of an artificial system? Pragmatic fixes have been proposed, such as the suggestion by Pearlman (2017) to treat AI outputs as works for hire in a corporate arrangement, sidestepping the need to determine whether the computer itself or the operator is the ultimate source of originality. Dornis (2020) argues that assigning protection to AI-generated works would incentivise productive investment in AI technology. Others have argued that it is a mistake to assign copyright ownership to expressions created by computers at all, fearing a proliferation of private rights and enclosure of the public domain as a result (Kop 2019). One source of anxiety is the potential for high volumes of expressions to be generated with minimal effort. Bridy (2012) makes reference to *The Great Automatic Grammatizator*, a short story by Roald Dahl (1954) which imagines negative impacts on creative markets from machines capable of generating limitless and personalised works, a plausible concern with the advent of generative AI.

Other than replacement by AI, some analyses locate human authors in networked relationships with AI as co-creators (Grimmelmann 2015; Lamerichs 2018; Craig and Kerr 2020; Hugenholtz and Quintais 2021). Lim (2018) suggests that working with AI might be analogous to a duet, with the AI interacting with artists ‘in a fashion like improvisational jazz, where musicians feed off cues from each other in a creative feedback loop’ (2018, 832). In this sense, AI-assisted creativity may be viewed as a pairing with, rather than a competitor to, human creativity, albeit with material effects. Foreseeing the use of AI as mediator between audiences and media products, Lamerichs (2018) suggests that creative fandoms ‘will be among the first social spaces with human and nonhuman participants’ (2018, 1). In her thought exercise involving a fictional TV series *Pleakwood Pines*, an AI system generates customised plotlines

that respond to audience tastes and inputs, yielding a networked platform experience more fully attuned to fans' desire for co-creation.

Co-productive possibilities of AI tools may conceal material effects on the experience of creative work. The use of AI as a production tool might exacerbate ongoing trends of deskilling and technological obsolescence, as for example by reducing the need for workforces in certain functions such as background music, translation or voice acting. In the case of music, Drott (2021) argues that the effects of AI have most concentrated in firms engaged in accompanimental music production, an already creatively marginalised sector. The effectively 'bottomless' capacity of AI to produce content may have the effect of saturating the market and undercutting human artists' revenues (2021). Farchy and Denis (2020) have explored the effects of AI in ancillary creative jobs such as A&R, marketing and regulatory compliance. They argue that AI is particularly attractive to creative industries due to their uncertain, superstar nature, where human judgment might be surpassed by an AI's ability to spot emerging trends (2020, 40). Widespread adoption of AI could prompt further movement away from publication models toward service-based offerings, further de-skilling labour and eroding professional incomes (Creus et al. 2020). In short, efficiencies enabled by adoption of AI across the creative value chain could have the effect of displacing human labour even if it does not fully replace core creative work.

2.2. Continuities: will AI accelerate displacement from digitalisation?

The effects of AI may be continuous with those already felt by creative workers as a result of digitalisation. Potential impacts include increased competition due to lower barriers to entry (Williams, McDonald, and Mayes 2021), changes in firm size and composition as a response to risk (Hesmondhalgh 2007), increase of outsourcing, freelance and short-term contracts (Creus et al. 2020), and oversupply of creative labour (Thomas, Battisti, and Kretschmer 2022). These factors are interrelated, with new technological developments like AI potentially playing a mediating or amplificatory role.

Creative production is thought to carry risk due to uncertainty about the reception of new products by audiences (Neff et al. 2005). Famously, 'nobody knows anything' until products are released to audiences (Goldman 1983 cited in Caves 2000, 3). While new technologies such as AI may mitigate these effects, the concentration of audiences for hits in digital markets suggests that management of risk continues to be an important factor in the failure or success of individual products and firms (Hesmondhalgh 2007). The capability of networked technologies to provide return-path information about consumer preferences is transforming management of digital creative industries in response to uncertainty (Schlesinger and Doyle 2015; Farchy and Denis 2020; Sotamaa, Tyni, and Myöhänen 2024).

Creative work is characterised by excessive labour supply and insufficient demand (Cunningham 2014). Many digital creatives report working multiple jobs to achieve stable income given the variability and uncertainty of creative work (Wright 2015; Thomas, Battisti, and Kretschmer 2022). AI adoption, like other processes of digitalisation, could exacerbate labour oversupply by providing new platforms and

technologies that lower barriers to entry and/or enable direct relationships between creators and audiences (Waldfoegel 2017; Williams, McDonald, and Mayes 2021). For example, more widespread use of software in music production and declining costs have given rise to a larger abundance of musical output and heralded the arrival of the bedroom producer (Rauh Ortega 2018). Similarly in photography, proliferation of digital equipment has lowered the skill barriers required to create images, while powerful editing software has made it easier for more people to cheaply create professional quality work (Williams, McDonald, and Mayes 2021). In response to increased competition, professional workers report pressure to continually maintain and refresh their technical skills or face obsolescence (Neff et al. 2005). New job types and identities have emerged in response to technological change, including the content creator and influencer roles, characterised by high levels of risk and pressure of constant professional self-invention (Banks and Deuze 2009; Duffy and Wissinger 2017; Williams, McDonald, and Mayes 2021).

Technology and sectoral developments have resulted in an overall tendency towards more outsourced and subcontracted work (Wright 2015). Workers have reported movement towards more flexible modes of employment, involving the need to be adaptable in terms of working hours, conditions, and contract duration (Banks 2010). Much work in the creative industries is project-based, involving short-term contracts and unstable working patterns with periods of low intensity and periods of 'crunch' (Creus et al. 2020). Banks and Hesmondhalgh (2009) have noted reduced bargaining power among creative industries and weak or absent trade unionisation in many places of work. Digitalisation has accentuated the outsourcing and flexibilisation of creative labour, for example over gig economy platforms in which creative labour is contracted on a task-by-task basis and workers are placed into competition directly with one another (Williams, McDonald, and Mayes 2021). Creative work happens in the context of a high proportion of small and micro-sized enterprises (SMEs). Similarly to the software industry, there is considerable churn in composition of firms, with startups forming and then splitting as employees start new companies (Doctorow 2014; Wright 2015).

Adoption of AI may further contribute to the autonomy/precarity dilemma in the characterisation of creative work (Gill and Pratt 2008; Banks 2010; Kovesi and Kern 2018). Despite unsteady and precarious employment conditions for many workers, creative jobs are often portrayed as 'cool' opportunities to exercise individual artistic autonomy (Neff et al. 2005; Banks 2010). From this perspective, it is almost inconceivable that AI could threaten demand for human labour. While conditions of work in the creative industries could be described as precarious and in a state of oversupply, the policy discourse frames these careers as exemplary opportunities to express individual (human) autonomy. They are seen as the opposite of routine 9-to-5 jobs that are presumably nearer the crosshairs of replacement by automation. In the traditional CI policy discourse, 'artistic-creative labour is concrete and named – it produces distinctive and 'authentic' commodities that cannot be subsumed into an impersonal or factory mode without losing the most significant aspect of their value' (Banks and Hesmondhalgh 2009, 416). At the same time, audiences would seem to prefer outputs linked to named and identifiable human creators (Banks 2010). Despite the seemingly resilient demand for human creative labour, inequalities in the system of creative

production may result in skewed benefits to fewer, privileged creative professionals. Widespread trends of deskilling taking place across the creative industries, accompanied by movement away from publishing models towards content (McKelvey and Hunt 2019) models of production suggest powerful counter currents eroding the working conditions and prospects of individual human creators.

As we have seen from the preceding discussion, there is evidence to support claims that AI will have negative effects for creative labour: either by substituting entirely for human artists, or by displacing human workers through processes similar to those already encountered from digitalisation. Despite the academic interest, largely from law and policy studies, on the potential impact of AI on creative industries, there is surprisingly little empirical evidence from the ground. More understanding is urgently needed to assess potential impacts from AI in real-world production contexts. The methodology outlined in the following section represents an attempt to add to empirical understanding of these effects, gathered through case studies with firms that have commercialised products using AI input.

3. Research methods

A case study approach was chosen to address the research questions: *How is the use of AI impacting creative work in firms that use the technology? Do we observe substitution or displacement taking place in this context? In what ways are any impacts continuous with ongoing challenges of digitalisation, and in what ways might they be discontinuous, requiring new policy focus?* Case study research involves in-depth study of a relatively small number of sites with the aim of understanding processes, systems or events. With a narrow focus on one or more bounded cases using multiple sources of data, it is an 'empirical inquiry that investigates a contemporary phenomenon within its real-life context' (Yin 2003, 13). Case studies are appropriate when research is exploratory, or when it covers a unique context where the approach can yield novel insights (Goffin et al. 2019, 595). This method is often used in research of live situations or processes for which experimental or other techniques are inappropriate, such as when the researcher is interested in human interaction and processes but does not have an ability to control or manipulate the environment (Ellinger, Watkins, and Marsick 2005). The case study method has been effective in studies of creative industries, notably in tracking external effects such as technological change or the funding environment on the prospects of creative producers and firms (Carpio, Birt, and Baumann 2023; Tan, Balaya, and Aaron Goh 2024). Case studies can offer higher external validity than other methods, due to the proximity of the researcher to practitioners and managerial situations in a given field. They be used for a variety of purposes, including for generating and testing theory (Voss, Tsikriktsis, and Frohlich 2002; Farquhar, Michels, and Robson 2020). In this research, the case study approach is applied to examine the impact of AI on creative work and assess whether its impact is continuous with established theory. I adopt a multiple case study approach to investigate whether differences exist across creative sectors and firm contexts.

To gain understanding of the use of AI in creative industries, this research applies a multiple, non-embedded case study approach, focusing on 6 creative products made

with the assistance of AI.³ The units of analysis in this study are bounded creative products, consisting of a single identifiable output generated in the context of a product team and involving at least one AI system in their production. Yin (2018) suggests that selection of case study sites should not seek to follow a sampling logic (e.g. with the goal of seeking generalisability) but should follow a replication logic (to generate new insights or test existing theory). In making the selection of cases, I follow a similar approach to Karjaluoto, Mustonen, and Ulkuniemi (2015) who chose an 'ensemble' of typical representative firms facing similar processes and using similar tools, in their case digital media communication technologies. Like those researchers, I also employ a comparative selection criterion to capture how companies are using AI in different organisational contexts. To provide a range of contexts, cases were purposefully chosen across different creative industries of publishing, music, computer games, performing arts, photography and architecture (see Table 1).

The main data collection method was semi-structured interviews carried out with lead creative personnel to obtain information about product development and creative decisions made when working with AI. Interviews were conducted following a 'funnel' protocol (Voss et al. 2005) with general questions about product design goals first, leading to specific probing questions about creative decisions at specific stages of production. Analysis of interviews followed a two-step process in which individual accounts were first coded to identify and interpret issues encountered when working with AI in specific contexts. In the second stage of analysis, categories and themes were developed by comparing experiences across individual cases. Within-method triangulation has been proposed as one way to improve internal validity and credibility of findings (Voss, Tsikriktsis, and Frohlich 2002). To enrich the cases and to help triangulate perspectives from informants, further data were collected and analysed from marketing materials, production notes, consumer reviews, and use of the products themselves.

Yin (2018) suggests that to improve construct validity, case studies should be designed to observe for specific concepts or operational measures. Key measures gathered and compared across cases include the type and size of underlying datasets used to train AI systems, the various forms of labour involved in production, and the extent of audience participation or co-creation in the output. Each of the main selected operational measures are shown in Table 1 below. Selected measures are summarised in columns d, e, f and g, and relate to amount of labour involved in projects both internal and external to the team, the amount and source of training data used in the project and the extent of any audience co-production in the project. A further criterion for comparison was the commercialisation strategy applied (as this reveals information about the role of AI and its potential future use by the firm). The next section summarises each of the cases and their production contexts.

3.1. Summary of cases

The 6 cases consisted of individual projects that were created with the assistance of AI. This section describes the production processes and organisational arrangements surrounding each project.

The Handsome One is an AI-produced chapter mimicking J.K. Rowling's writing style, following the official Harry Potter books. The producers, Botnik Studios describe

Table 1. Summary of case studies and main operational measures.

A. Product name	B. Date of initial release	C. Organisation / Firm	D. Internal team size	E. Dataset(s) used	F. Audience co-creation	G. Commercial strategy
The Handsome One	December 2017	Botnik Studios	4 core members of firm plus several contract writers	Text from the 7 canonical Harry Potter books	High. Up to 200 community contributors	The product is a free demonstration of a predictive text platform and consulting the company sells
Play Narrator	April 2023	Dave's Games	1 sole operator	Multiple open-source and proprietary Large Language Models (LLMs) used interchangeably	Medium. Users can create and publish their own game worlds as UGC on the platform, and feed into development via Discord	Subscription service via web platform, with limited free-to-play version
The Living Archive	March 2021	Wayne McGregor Studio with Google Arts & Culture Lab	1 main creator and 3-5 contract employees from Google	Several hundred hours of video-recorded dance performances	Limited. Users can interact with online tool but licensing prevents further use	Ticketed performances based on the AI output. A free online platform was also published.
Beautiful the World	May 2020	Uncanny Valley	8 production staff, some on contract	200 Previous Eurovision Song Contest entries; audio samples of wildlife	Limited. Voting by anonymous audience members 2 wk prior to competition	Competition entry. Platform is used as demonstration for an AI product the company sells
Generated Humans	December 2022	Generated Photos	20 employees plus contracted studio photographer	Proprietary set of 70,000+ photos of human models	Limited. Client feedback shapes evolving product offer	Sold as a subscription API platform or via bespoke commissions
Argenia architecture generator	1986, upgraded intermittently	Celestino Soddu, Architect	1 main designer with colleagues and technical assistance	Archive of own work, previously generated designs	Intermittent, via norms of academic exchange	Demonstration as part of the architectural firm's overall portfolio of services

themselves as ‘a community of writers, artists and developers building and using machine tools to remix and transform language’ (Botnik Studios 2021). The studio consists of a core team of paid employees and a wider community of approximately 200 users and contributors. The company views its use of AI tools as collaborative: ‘we would like, selfishly, not to replace humanity with algorithms. Instead, we want to find natural ways for people and machines to interact to create what neither would have created alone’ (Botnik Studios 2021). *The Handsome One* was produced using a combination of AI predictive text and human curation. The core team organised a writing jam in which contributors generated text from a dataset of the 7 canonical Harry Potter books. Producer Nat Towsen then ‘took the best pieces of copy, arranged them into a narrative, and wrote some copy to fill in the gaps’ (Towsen 2017). *The Handsome One* has not been directly commercialised, but helped to raise the profile of the company when it was featured in various online and offline press, including the BBC and Newsweek. The company earns revenue by offering consulting services based around computer-generated text. These include experiential workshops offered to larger companies.

Play Narrator is a text-based computer game product that enables players to create their own character and companions and explore a fictional text adventure generated by an AI. The product uses interchangeable LLMs such as GPT3.5 to play the role of ‘Dungeon Master’ and present the player with both a textual description of the world as well as the outcomes of the player’s actions. Advertising itself as offering ‘any story, any game’, the platform features open-ended and user-generated game scenarios across a range of categories. The core of the game was developed by designer David Tompkins over a 2-month period in 2023 and has been continuously upgraded with new functionality, including the ability to generate images depicting the player and their adventures. The product is offered for free on the company’s website, with a premium subscription model offering players access to more powerful models and more functionality.

The Living Archive is a collection of works that include dance performances and an accompanying online tool, co-produced by dance choreographer Wayne McGregor and Google Arts and Culture Lab. The project, initiated by Wayne McGregor Studio used a dataset of several hundred hours of video spanning the choreographer’s 25-year career and drawn from rehearsals and performances. The video data were parsed using a ‘pose-detection’ algorithm by Google Arts and Culture Lab team, and then an AI system was used to generate predictive dance moves or phrases from the sequence of existing poses. A dance performance based on the collaborative work, titled *Living Archive: a performance experiment* was commissioned and performed at LA Philharmonic in July 2019. Promotional material described how the AI system would ‘respond to any new movement phrase by suggesting multiple original possibilities for the next phrase, [...] bringing dancers of the present into contact with traces of their predecessors’ (Wayne McGregor Studio 2019).

Beautiful the World is a pop song created by music production studio Uncanny Valley for the 2020 AI Eurovision Song Contest, organised by Dutch broadcaster VPRO. Uncanny Valley is an Australian music production company founded in 2010 by Justin Shave and Charlton Hill. The company describes their emphasis on

computer tools in music production as ‘augmented creativity’ (Jewell, 2021). The main business activities of the studio are work-for-hire music production, including ‘sonic branding’ for television formats, musical direction, sound design and mixing. Two datasets were used to train the AI systems in *Beautiful the World*. One consisted of sound recordings of 200 previous Eurovision Song Contest entries, supplied under license by the event organisers as part of the terms of participation in the contest. A second proprietary dataset consisting of audio recordings of Australian wildlife was obtained by uncanny Valley as part of their creative choice for the melody, which they envisioned ‘as a response to the huge forest fires that devastated Australia [...] and which took the lives of many animals’ (VPRO 2020). Subsequent to the AI song contest, the company developed a prototype AI music platform called MEMU which can generate live music based on inputs of samples of other songs. The prototype demo incorporates the dataset of Eurovision song contest melodies employed in the creation of *Beautiful the World* and is used to promote Uncanny Valley’s commercial services.

Generated Humans is a bulk photography product that offers artificial portraits of human faces generated by AI. The firm, Generated Photos is incorporated in the USA but describes itself as a ‘fully distributed company’ with 20 employees worldwide (Generated Photos 2023). According to the firm, the primary enterprise customers for the product are video game development studios, film and television production companies, and other technology firms working on machine vision. The product is marketed as ‘worry free,’ due to the lack of a requirement to obtain model release permissions for non-human subjects. The initial dataset consisted of 70,000 photos of human faces made by a contracted studio photographer, with additional photographs added over time to increase diversity and accuracy. The product is commercialised *via* subscription to an API that enables customers to generate portraits according to customisable criteria. Bespoke orders are also negotiated with large clients on an individual basis. Since commercialisation of the first-generation product, the company has expanded its offering to include full-body human models, with a similar commercialisation strategy.

Argenia is a proprietary generative system that produces 3D architectural forms, based on inputs from the creator’s previous work. The project was developed by Italian architect Celestino Soddu in 1986, with continual updates. *Argenia* differs from other cases in that it does not use machine learning techniques, which have come into more widespread use relatively recently. The system as described is closer to a generative algorithm, which automatically generates and then builds upon previous iterations of architectural plans with supervision from the human operator. The lead architect stated that as a result of this supervision, the system is ‘able to generate and endless sequence of different 3D models of industrial objects, all unique and unrepeatable, but all belonging to the same design concept’ (product developer 6). The product was not directly commercialised, but was used to support the architectural business as an innovation showcase and part of the firm’s overall creative offering. *Argenia* was also used by the creator in his teaching as professor of Architecture at Milan Polytechnic University, where the product was also further developed with input from students and colleagues.

4. Discussion: working with AI

Product teams and individuals reported different experiences working with AI, depending on their function within the overall production process. In each case, participants described an experimental approach, foregrounding learning, iterative development and adaptability. Consistent with trends across the wider creative sector, the majority of the projects involved geographically distributed teams, with some contributors working on a short-term contract basis. In general, working conditions were not markedly different from similarly positioned digital creative firms. However, the case studies identified characteristics specific to work with AI. Overall, the case studies highlighted: (1) extensive labour involved in AI production, which included all of the work normally associated with traditional media production, plus additional expertise in computer software and data curation; (2) the collaborative nature of production involving AI, including co-production by audiences and contributors from beyond the firm; (3) the tendency for human labour to be invisibilised, even when extensively present, in marketing and other discussions about products; (4) simultaneously, audience awareness and fascination with the unique characteristics and imperfections of AI outputs, complicating the substitutability of AI products for human creations.

4.1. Significant investments of labour

In general, respondents characterised working with AI in creative roles as highly labour intensive. Nearly all of the products studied involved traditional creative production work as well as new, non-traditional labour and skills. Much of the new labour related to selecting and preparing data for input, as well as tweaking and curating output from the AI systems. Several respondents reported that they were required to learn new skills as part of their contribution to AI product development. When asked whether the use of AI offered any shortcuts or time-saving in music production, the lead developer for *Beautiful the World* disagreed:

It was definitely so much more work. Normally, one wouldn't be trying to develop a whole high-tech system when you're writing a song. Well certainly in my case, my contribution was kind of like a developer more than a creative for that project. I mean I did use my own knowledge of song-writing to guide the ideas and the contributions that I made, but ultimately I was not the person putting the final pieces together to create the song.
(Product developer 4)

In this case, a programmer's work on the project involved unique overlapping skillsets as a software developer and as a musician, enabling her to contribute productively across both domains. Considerable post-production was also involved in creation of the contest entry, comparable to that of a traditional commercial pop song. The lead developer described how the final output or 'render' of the song was not fully automated by the computer. The Uncanny Valley team later 'relied on a human singer to provide backup vocals, resulting that the song was 'a blending of post-processed Sinsy with the lead singer Antigone' (product developer 4). Other cases also reported significant investments of human labour to conceive and design the overall project, even when automation added efficiencies in other areas. The developer of *Play Narrator* reported spending 2 months working on a prototype of the game, after improving

their coding skills by building other smaller projects. The majority of the work to develop *Play Narrator* was described as integrative, taking blocks of code developed with the help of AI to build a working system integrating different models for different purposes. The developer also expressed significant time spent maintaining and upgrading the product after launch. This involved adding new models to the game and troubleshooting issues encountered by users. One of the developers of *The Living Archive* choreography project described that the bulk of the work involved preparing data for the purpose of AI training:

A lot of the work was actually in the cleaning of the data. We had lots and lots of detected poses from the database. But the detection algorithm doesn't work in all cases. [...] There were a lot of video interviews, things like that. So quite a bit of work went into cleaning up the data for later training the model. And also tweaking the model and parameters to make it work. (AI collaborator 3)

While AI would later be able to introduce efficiencies by swiftly analysing the trove of choreography videos for insights to predict and generate new choreography, the lead developer explained that time-savings 'was not really the point'. The objective was 'to explore what a machine can do and what it can propose, in order to inspire new movements that a human wouldn't have thought of... achieving that is impossible by definition to do with a human brain' (AI collaborator 3). Even when efficiency was the stated goal, such as in the *Generated Humans* bulk photography product, the firm described painstaking and repeated work by a traditional studio photographer and machine learning experts to capture and then improve training data to feed the system. When feedback came from clients that the generated portraits were 'too attractive' for certain use-cases, the photographer was once again employed to repeat the process over again with a more diverse set of models (product developer 5).

Where labour savings was reported, this occurred after the difficult work of training and refining the AI system, and was primarily used in the ideation phase. Architect Soddu reported that with the current iteration of *Argentina*, he could 'generate an STL file of a 3D model of about a million faces' in 2-3 min, that could be converted into a physical model using a 3D printer (product developer 6). This enabled him to explore new design possibilities swiftly, avoiding work that might previously have been done by human assistants. One Google Arts Lab member imagined similar possibilities for choreography, whereby 'instead of having the [human] dancer do all of the variations they can think of, we say to the computer, OK, start with this movement and give me thirty options on how to complete that movement' (AI collaborator 3). However, in order to remain useful as an ideation tool, AI systems and datasets would require continuous updating and refinement to incorporate changes to the creators' style.

4.2. Co-production, collaboration, iteration

Many of the products involved collaboration with external contributors and followed iterative product development cycles with significant experimentation and testing. The AI products studied here did not arrive to market fully formed; rather products were refined over time with significant input from a variety of creative sources and

were rarely dependent on a sole human author or team. Two firms maintained crowd-sourcing communities as part of their production process. *The Handsome One* Harry Potter chapter was created with direct input from collaborating users in a writing jam format, using the firm's text generation system to suggest short excerpts. The producer described how he then selected and later stitched together 'the best pieces' from the various human participants in the activity (product developer 1). The *Play Narrator* website provides a platform for users to share and play each other's games, which consist of artwork, characters and world parameters. User-created games are showcased on the main page, along with a prominent 'Create' button. The developer also offers a Discord server where players can discuss their experiences with the software. In both cases, products potentially benefitted from users' volunteer labour in a 'private-collective' innovation arrangement (Von Hippel and Von Krogh 2003). Other firms took a more traditional approach to product development, maintaining more internal control over the process but nevertheless relying on iterative testing and refinement. Architect Soddu described his *Argenia* product as an accretion of his own ideas and work spanning over 40 years, with new adjustments and additions to the algorithm made as his style and influences changed, sometimes with input from others. The product team for *Generated Humans* similarly described a process of constant refinement, iteration and experimentation driven by requests from clients:

It's a constant process because there are so many things to improve. We keep getting feedback and we see space for improvement. The more data you have the better. If you want to feature something new, it requires more data. [...] Let's say you want it to recognize or to generate images of Pacific Islanders. For that you need lots of images of Pacific Islanders, so that it knows how they look like. As I mentioned we are working on full-body generation. We are not there yet, but we have some promising results. And that's a big step forward. (Product developer 5).

The amount of labour, distributed in both time and space in the development of these AI products suggests that the *Automatic Grammatizator* (Dahl 1954 cited in Bridy 2012) is – at least for now, in this experimental context – an inaccurate characterisation. Continuous with (Sotamaa, Tyni, and Myöhänen 2024) whose respondents describe 'wrestling with' algorithmic systems, it is not a case of pressing a button to obtain an instantaneous output, but rather a prolonged and sometimes difficult engagement with technology involving a range of human labour, both credited and un-credited and involving paid as well as volunteer labour in certain cases.

4.3. Marketing and audience perceptions: replacing the human?

As discussed above, some have expressed concern that AI products may compete directly with human-made outputs on the basis of quality, and out-compete human authors in terms of quantity (Kaminski 2017; Brown 2018; Kop 2019). The products analysed here were marketed in such a way that the contribution of AI systems was prominently foregrounded, sometimes to the point of invisibilising human labour that went into their creation. *Play Narrator* is marketed as an 'infinite AI adventure role-playing game'. The website is presented in a minimalist style, with the FAQ inviting users to experiment and 'lean-in' to interacting with the AI models powering the game. In press materials covering *Beautiful the World's* success in Eurovision AI Song

Contest, it was stated that ‘AI was [...] solely responsible for creating both the lyrics and the melody’, over-simplifying the curatorial and guiding role of humans throughout the process (Jeffery 2020). Uncanny Valley members were featured in the reporting, but their faces were obscured in a manner reminiscent of Daft Punk.

Despite the foregrounding of technology in the marketing of AI products, developers were not convinced that their outputs would be seen as a substitute for adjacent work produced by humans. As one developer put it their project was ‘*about* AI, not only made with AI’ (AI collaborator 4). When designing and selecting how to present AI outputs, some of the teams focused on making obvious the differences between human and computer creativity. Audiences were invited to investigate the unique qualities of the AI contribution to the works and the new perspectives that offered on the underlying art form or genre, such as the Harry Potter books used in *The Handsome One*:

We tend to play with texts and genres that people are deeply familiar with, so when our algorithmic approach yields something that’s wrong, we know immediately that it’s wrong, and there’s a sort of joy in realizing how well you know this book, or this genre, such that you can tell immediately when something does or doesn’t make sense. (Martin 2018).

Creative managers explained that they were cognisant of audience expectations as part of their creative process, aiming to highlight unexpected differences between human and computer expression. The production team of *Beautiful the World* purposefully selected aspects of the computer output that seemed uncanny and amplified them in the final production. The music programmer for the project was hopeful about the potential for her system’s awkwardly-generated lyrics to circulate as memes: ‘If this song becomes popular, I can see this style of communication catching on in a similar way to LOLcat phrases (‘serious cat is serious’), and other awkward phrasing’ (product developer 4). In the case of *The Living Archive*, one programmer described how he found creative inspiration from the bugs and glitches produced by the AI system, and viewed those as the most interesting aspects of the performance. He was disappointed by some critics who he felt missed the point of the performance being about AI-human collaboration rather than replacement of the human creator:

Some of [the responses] were from classical dance reviewers. And most of them in my opinion missed the point. They would say stuff like “OK, you know, computers will never replace humans, and they have no sensitivity” and stuff like that. And my reaction to this is “yeah, of course, this is the point”. It’s trying to see the collaboration, and what [working with an AI system] can bring that is important. (AI collaborator 3).

In the way that designers hoped to use AI as a means of exploring the possibilities of the technology and reflecting a mirror back onto established artistic norms and practices, some of the AI products more closely resembled fan fiction. For that reason, the developer of *Play Narrator* felt that their AI game platform would not directly compete with commercial video game releases, because players were more interested exploring personal stories for which there was not a sufficiently large commercial market.

[O]n my game, everything that’s created that way is kind of ephemeral. Like, it’s not really replacing a human. If it wasn’t for AI models, those stories just wouldn’t exist. Like, you

know, no one would hand paint this character for your own private game. It would be very cool if they did, but that would be really expensive. So, I don't feel like it's taking anything away [from human artists] there. (Product developer 2)

Like human storytelling and creative pursuits, AI outputs relate to 'a vast sea of texts' that comprise the training data for these systems (Craig and Kerr 2020). The relational nature of both human as well as AI-assisted creativity further complicates the role of the human author and introduces legal uncertainties that constrain commercialisation.

4.4. Risk and commercialisation strategy

Working with AI presented individual as well as commercial risks. Some of those involved in production of the case study products were hired on a contract basis and were not retained after the completion of the project. These were the AI and computer specialists on *Beautiful the World* and *Living Archive* projects. In both cases, the AI specialists moved on to other paid projects, while the core creative teams in firms continued to work in their previous creative roles, not necessarily involving AI.

Success in commercialisation varied from case to case. As previously observed by Drott (2021) in the music sector, a number of cases adopted platform models of commercialisation, at least in part. Botnik Studios offers a free interactive demonstration of their predictive text tool called Voicebox, which also serves as a demonstration for attracting commercial consulting work (Towsen 2019). One of the outputs of the *Living Archive* is an online interactive tool that allows users to visualise poses detected by the AI and join them together to create new choreographies. *Play Narrator* is offered in a free trial version as well as a paid subscription service via the company's website. The developer reported that the platform was breaking even in terms of profitability, with the main cost being access to AI models. They had used the experience developing the game to gain contract work on related AI projects. In general, these free AI product demonstrators served as complements to other paid products or consulting work offered by the firms involved. The diverse approaches highlight that commercialisation success is highly uncertain, leading firms to experiment with a variety of revenue models.

Direct commercialisation can be challenging due to the intellectual property status of upstream training data used in some cases (also observed by Drott 2021). The makers of *Beautiful the World* had permission from the rights-holding broadcaster to make use of the database of songs, provided under license as part of their participation in the contest. They were required under the license not to share or distribute these data which were for the exclusive use of competition participants. While Google Arts and Culture Lab had permission from Wayne McGregor to use his choreographed dance performances in the archive for pose detection, neither of the collaborating parties owned the rights to the totality of videos in the archive, many of which were owned by various venues around the world where the recordings were made. The inability to clear the rights in all video material made it impossible for the collaborators to include or use video output in the final product. The developer of *Play Narrator* provides a DMCA takedown procedure on their website for rightsholders seeking to remove copyright-infringing material, although the company reported

never having received a request. Removal or changes to AI model access terms was a risk that the company dealt with by swapping to alternative model providers, although this required maintenance work. When using upstream training data in which the IP permissions are ambiguous, smaller firms might hope to pass ‘under the radar’ of incumbent firms, gain experience, attract a customer base, and then, in a second phase, engage in commercialisation (Haeffliger, Jäger, and Von Krogh 2010: 1198). However, this approach carries significant risk, amplified by working with AI where significant up-front investments must be made in obtaining training data or developing products for use with specific models.

5. Conclusions

The case studies revealed that firms understood working with AI as a co-creative process (Lim 2018) rather than as a replacement for human artists. Producers viewed the technology positively and emphasised that it allowed them to explore new kinds of creative expression rapidly and with fewer limits than traditional methods of ideation. Firms adopted an experimental and iterative approach, similar to existing software development practice. This involved significant amounts of co-creation with users and audiences during the development process. The open, iterative and experimental approach to working with AI led to commercialisation challenges. Platform subscription models allowed for continuous improvement and iteration but required upkeep and management. Most participants viewed the AI products not as direct sources of profit, but as complementary to other offerings or a way to gain future work. Production using AI mirrored many of the trends observed in digitalisation, such as venture labour and upskilling, although case study informants described these in terms of curiosity and exploration rather than structurally imposed.

Challenges faced by creative SMEs from digitalisation were also encountered by the case study firms. Product teams were small, flexible, distributed and subject to considerable risk, all characteristics identified in previous studies of digital creative SMEs. Case study firms embraced AI because it offered new possibilities for creative expression, but this brought new complexities and knowledge requirements, often requiring collaboration across sectors or beyond the boundaries of the firm. Working with AI was experimental, exploratory, and introspective. Case study firms used AI as a means of exploring the interface between computer and human creativity, and to reflect back like a mirror on the styles, conventions and genres of previous creative practice underlying their artforms.

There was limited evidence of displacement of creative work inside of the case study firms, where implementing AI required more work rather than less. However, the case studies suggest hypothetical competitive effects from more widespread adoption of AI across the industry. The *Generated Photos* product likely competes with other, older methods of obtaining large quantities of licensed imagery, such as stock photography providers and modelling agencies. Architectural generation systems like *Argenia* could potentially displace work previously done by assistants if adopted by larger architectural firms. By providing a limitless sandbox to explore fictional stories, *Play Narrator* potentially competes with other text-based adventure games. For the

most part, the products undertaken by the case study firms were one-off experiments, whose effects remain limited to the context of the firm developing and using the technology. One avenue for future research is to explore how other types of firms that would normally hire creative freelancers are making use of low-barrier-to-entry AI platforms like DALL-E and ChatGPT, and how the widespread availability of such platforms is being encountered across the sector.

New risks for creative firms using AI include ambiguity about the legal status of upstream datasets and AI outputs. Some of the respondents reported working in a legal grey area. Ingesting, curating, cleaning and preparing data were reported as significant tasks for human labour in AI projects, and this investment of resources carried significant risk if products failed or required re-design. Legal certainty about the status of upstream datasets emerged as a key component of the ability of firms to commercialise AI products based on those training inputs.

Policy makers and researchers should be attentive to the relational, collaborative, and iterative nature of AI production, and particularly the ways in which upstream datasets and downstream audiences are co-implicated in these processes. These features of AI production mirror processes already observed in digitalisation, such as the increased role of audiences, fans and users in shaping creative production. The case studies indicate that AI-assisted products are highly complex from the point of view of IP licensing and ownership. Somewhat like the early days of the internet, a lack of coordinated regulation of AI creativity seems to be enabling significant technological and legal experimentation, along with accompanying risk to participants and firms. And like the commercial expansion of the Internet in the late 1990s, the arrival of larger corporate players, investors and regulators will likely impact the landscape for creative work.

Notes

1. A test, conceived by Alan Turing in 1950 to establish the apparent sentience of a computer system. The test would involve an observer tasked with determining the status of human and computer conversational partners whose identities were concealed, based solely on the responses received. See: <https://plato.stanford.edu/entries/turing-test/>.
2. In reference to the 19th Century mathematician's comments on human perceptions of machine potential.
3. For the purposes of this study, "AI-assisted media" are defined as media created with the help of computer techniques that mimic human learning or decision-making. Common techniques include predictive algorithms, machine learning (ML), generative adversarial networks (GAN) and recurrent neural networks (RNN).

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